

Caveat Emptor

Recent cases reveal that the efficiency of a product is sometimes less than meets the eye.

In 2006, *Consumer Reports* tested the LG Energy Star bottom freezer refrigerator, the LG LRFC25750, and discovered that it won't meet Energy Star standards in most homes. (See *Consumer Reports*, August '06, p. 8.) DOE doesn't normally test appliances; instead, it relies on manufacturers to test and report on the energy use of their products. The DOE test calls for the manufacturer to measure refrigerator efficiency in a chamber set at 90°F with the refrigerator doors closed. That's not how most refrigerators really operate, but the losses from the higher ambient temperature are supposed to compensate for doors opening.

Consumer Reports tested the LG refrigerator under a variety of ambient temperatures with the door opened and closed and made a surprising discovery. An electronic sensor in the refrigerator regulates a heater that is used to keep the door gaskets pliable so that the gaskets provide an effective seal around the refrigerator doors. But the sensor turns off the heater when the ambient temperature reaches 90°F. With the 100W heater on, the refrigerator energy use exceeds the maximum allowed for Energy Star designation. (The LRFC25750 is rated at 499 kWh per year. The *Consumer Reports* article does not give the tested energy use, but one consumer reported measured energy use of nearly 700 kWh per year for a similar Energy Star-rated LG refrigerator rated at 465 kWh per year.) With the heater off, the refrigerator meets the Energy Star standard. In other words, the DOE test will always pass the refrigerators. But in most homes, where average kitchen ambient air temperatures are close to 70°F, the refrigerators will use much more energy.

And in Australia, a different LG home appliance has come under scrutiny. The company has been fined



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more than \$2.3 million because five air conditioner models did not meet stated cooling output levels, used more energy than the rated consumption, or failed to meet Australia's minimum energy performance standards (MEPS). The Department of Energy, Utilities and Sustainability in New South Wales, and Energy Safe Victoria tested the five models. The money collected in fines will be given to the buyers of more than 15,000 mislabeled LG air conditioners to compensate them for unexpected energy use.

The recent compliance problems are not limited to appliances. The Federal Trade Commission (FTC) has fined an insulation manufacturer for wrongful R-value claims. "Insulation manufacturers should be familiar with the requirements of R-value rules," says Hampton Newsome, an attorney with the FTC's Bureau of Consumer Protection. "The FTC sets standards for testing, labeling, and the disclosure of thermal performance," says Newsome, to give consumers an objective measure

of insulation quality and to allow consumers to compare one insulation product to another. The FTC levied a fine of more than \$100,000 against Northwestern Ohio Foam Packaging, Incorporated, and its owner, Wally Radjenovic. In its advertising, the company claimed an R-value for one product "The Barrier," that was 6 times the R-value found through testing. The Barrier is an under-concrete thermal insulation and vapor retarder with a core made of extruded expanded polystyrene (EPS) and 3-mil poly film laminated on both sides. The company was also charged with not labeling the R-value on The Barrier and its Microfoil radiant barrier product, and for breaking other R-value rules.

Even a simple showerhead may be out of compliance. In December 2005, the city of Seattle and the California Urban Water Conservation Council (CUWCC) petitioned DOE to bring sanctions against the manufacturers of water-gushing showerheads. "We were horrified to find showerheads for sale in California that flowed three to five times above the legal limit," says Mary Ann Dickinson, executive director for the CUWCC. Federal law sets a minimum flow rate of 2.5 gallons per minute (gpm) for new showerheads, and it is against the law in California and Washington to install higher-flow showerheads. But independent testing of the suspect showerheads at the request of the city of Seattle uncovered flow rates of 7.6–13 gpm. And as of this writing, there are still gushers available to buy off the Web. Just Google "showerheads 10 gpm" and see what comes up. (See "Compliance: Following the Letter (and the Spirit) of the Law," p. 2.)



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